

ORANGEBURG, SC, USA

WMO: 723115

Lat: 33.462N

Lon: 80.858W

Elev: 197

StdP: 14.59

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 53854

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.9	28.0	6.8	7.9	34.2	11.2	9.8	38.7	22.6	56.8	19.9	56.2	3.3	10	0.383

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	96.0	75.9	93.4	75.5	91.4	75.1	79.0	89.8	78.0	88.5	77.2	87.2	8.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.2	137.5	82.9	75.2	133.1	81.9	74.4	129.6	81.1	42.5	89.5	41.4	88.8	40.6	87.4	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.6	16.3	13.7	DB	19.6	98.9	3.6	2.6	17.0	100.7	14.9	102.2	12.9	103.7	10.2	105.5
			WB	17.1	80.6	3.1	1.4	14.9	81.6	13.1	82.5	11.4	83.3	9.2	84.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.0	47.0	50.4	56.9	64.7	72.2	78.9	81.4	80.8	75.8	65.8	55.6	49.8
	DBStd	14.15	10.05	8.77	8.61	7.05	5.63	3.83	3.16	3.29	4.90	7.42	8.17	9.21
	HDD50	471	183	95	35	2	0	0	0	0	0	2	34	120
	HDD65	2212	563	415	275	91	12	0	0	0	2	81	296	477
	CDD50	5950	88	105	248	442	688	868	973	954	773	492	204	115
	CDD65	2216	3	6	23	81	235	418	508	489	325	106	16	6
	CDH74	20392	16	41	235	825	2179	3952	5081	4577	2563	792	108	23
	CDH80	8599	0	2	35	213	815	1818	2449	2075	998	187	7	0
(14) Wind	WSAvg	5.6	6.3	6.5	6.8	6.6	5.8	5.2	4.8	4.5	5.0	5.0	5.1	5.7
(15) Precipitation	PrecAvg	47.7	4.3	3.8	4.4	3.0	3.7	5.0	5.7	5.3	3.9	3.0	2.6	3.1
	PrecMax	71.5	8.7	7.9	10.2	10.7	9.8	10.9	13.9	13.5	12.0	17.4	10.2	6.7
	PrecMin	31.2	0.6	1.1	0.7	0.2	0.0	0.7	1.6	0.9	0.0	0.0	0.2	0.4
	PrecStd	8.6	1.9	1.7	2.3	2.0	2.4	2.3	3.0	2.8	2.5	3.0	1.9	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.4	79.1	83.9	88.4	94.8	98.1	98.9	97.8	93.9	88.2	81.2	77.3
		MCWB	64.3	64.6	64.1	67.4	72.1	75.1	76.7	78.0	72.9	73.1	67.5	66.8
	2%	DB	71.6	74.1	80.2	85.3	90.6	94.6	96.3	94.4	90.9	84.6	77.3	72.9
		MCWB	61.4	61.8	63.4	66.7	70.7	75.0	76.6	76.8	73.5	70.2	64.8	64.4
	5%	DB	67.4	70.4	76.3	82.2	87.9	91.8	93.3	92.1	88.7	81.7	73.4	69.3
		MCWB	59.3	58.7	61.1	65.6	70.0	74.4	76.3	76.2	72.9	68.3	62.4	62.8
	10%	DB	63.4	66.0	72.4	79.2	84.4	89.7	91.0	90.2	86.2	78.7	70.3	65.7
		MCWB	56.8	57.2	59.4	64.1	69.2	73.8	76.1	75.7	72.5	67.3	62.1	60.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.7	67.0	68.1	72.3	75.4	78.5	80.8	80.5	77.6	76.1	71.1	70.0
		MCDB	71.8	74.5	77.2	81.4	86.3	89.9	92.1	91.6	86.0	82.5	75.6	74.0
	2%	WB	64.3	64.6	66.1	70.3	73.7	77.4	79.1	79.2	76.5	74.3	68.6	66.8
		MCDB	69.6	71.3	74.6	79.4	83.6	89.0	90.4	89.6	84.6	80.2	72.9	70.7
	5%	WB	61.6	62.0	64.3	68.6	72.5	76.4	78.1	78.1	75.5	72.5	66.4	64.1
		MCDB	66.0	68.1	71.9	77.5	82.1	87.1	89.2	88.0	83.3	78.2	71.1	67.8
	10%	WB	58.3	58.8	62.3	66.7	71.2	75.4	77.2	77.2	74.5	70.4	63.7	60.9
		MCDB	62.3	64.3	69.4	75.2	80.5	85.2	87.8	86.4	81.6	75.7	68.9	65.5
(35) Mean Daily Temperature Range	5% DB	MDBR	20.5	21.6	22.8	23.4	21.9	19.7	19.1	18.1	18.5	21.1	22.6	20.3
		MCDBR	24.1	26.0	26.8	26.8	25.7	23.1	22.0	21.0	21.8	22.7	25.0	22.8
		MCWBR	15.3	15.4	12.6	11.2	8.9	7.3	6.8	6.5	7.4	10.2	14.1	15.7
	5% WB	MCDBR	19.7	22.7	22.2	21.2	20.4	19.5	19.6	18.7	17.5	16.8	18.6	19.4
		MCWBR	14.6	15.7	12.7	10.6	8.0	7.1	7.1	6.6	6.8	9.0	12.8	15.3
(40) Clear-Sky Solar Irradiance	taub		0.325	0.333	0.369	0.405	0.474	0.512	0.552	0.523	0.437	0.402	0.348	0.321
	taud		2.492	2.464	2.391	2.283	2.135	2.072	1.983	2.055	2.285	2.345	2.464	2.539
	Ebn at Noon		282	291	288	280	261	250	239	244	262	263	271	277
	Edh at Noon		27	31	36	42	49	52	57	52	40	35	28	25
(44) All-Sky Solar Radiation	RadAvg		898	1098	1440	1817	1960	2019	1940	1757	1514	1280	986	790
	RadStd		47	111	109	124	145	140	131	92	131	152	75	71

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.06	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)		+1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+347	+243

Nomenclature: See separate page